Shubham Vetal

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Robotic Systems Development (MRSD) — May 2028
Current Coursework: Robot Learning, Robot Mobility, Robot Autonomy, Systems Engineering

Pune Institute of Computer Technology — Pune, India
Bachelor of Engineering in Electronics and Telecommunication Engineering — June 2024
GPA: 9.32/10.00

SKILLS

Programming Languages: *Advanced* - C++, Python; *Intermediate* - C
Robotics: ROS, ROS 2, Robot Perception & Control, OpenCV, MoveIt, Nav2, Behavior Trees
Simulation & CAD: NVIDIA Isaac Sim, MuJoCo, Gazebo, MATLAB, Fusion 360
Tools & Environments: Linux (Ubuntu), Git, Docker

PROFESSIONAL EXPERIENCE

Addverb Technologies Limited — Noida, India
Robotics Engineer — December 2024 - April 2026

- Developed ROS-based perception and autonomous docking for AMR forklifts, achieving ±5 mm docking accuracy and robust detection across ±30° at 2.0–2.5 m.
- Deployed a 13-robot autonomous forklift fleet, achieving 30+ picks and drops per hour per robot.
- Enabled the transition from LiDAR-centric to vision-based perception, reducing system cost by 8–10% while improving reliability. Contributed to the “AMR Forklift – Pallet Docking” patent filing.

Addverb Technologies Limited — Noida, India
Mobile Robotics Intern — June 2024 - November 2024

- Benchmarked ORB-SLAM3, RTAB-Map, and SVO on AMRs to characterize localization and runtime trade-offs.
- Evaluated SE(3)-TrackNet and FoundationPose to determine the feasibility of one-shot 6DoF pose estimation.
- Engineered homography-based obstacle detection and ICP-based pallet-localization workflows.

SoulEET Makers LLP — Pune, India
PCB & Embedded Hardware Design Intern — October 2023 - March 2024

- Designed and optimized a multi-interface ESP32/SIM7600 PCB with Ethernet, RTC, and I²C/SPI/UART interfaces, improving battery backup by 20% and output efficiency by 10–15%.
- Owned PCB development from component selection to fabrication and deployment, delivering a GSM monitoring system that reduced manual intervention by ~80% with long-term field reliability.

ACADEMIC PROJECTS

Mecanum Drive Robotic Arm Manipulator — Pune, India
Pune Institute of Computer Technology — July 2023 - March 2024

- Developed a ROS 2 mobile manipulator with Raspberry Pi, LiDAR, camera and IMU for warehouse tasks.
- Leveraged SLAM Toolbox for mapping, Nav2 for autonomous navigation, and MoveIt for manipulator motion planning; validated the system in Gazebo simulation and deployed it on the physical robot.
- Integrated monocular-camera ArUco detection for inventory identification and material labeling. Won First Prize at Instructables’ “All Things Pi” contest.

IoT-Based Smart Security Surveillance Robot — Pune, India
Pune Institute of Computer Technology — January 2022 - June 2022

- Created a Raspberry Pi surveillance robot with Wi-Fi control, IR obstacle sensing, and 180° video streaming.
- Integrated web teleoperation, camera control, and real-time sensor visualization for remote security monitoring.
- Deployed an end-to-end Flask web-control system. Won the Judges Prize at Instructables’ Electronics Contest.

LEADERSHIP & INDUSTRY ENGAGEMENT

AMR Deployment Representative, Addverb Experience Centre - Zoetermeer, Netherlands — April 2025
Product Launch Representative (LogiMAT), Addverb - Stuttgart, Germany — March 2025
Team Captain, PICT Robotics - Pune, India — September 2022 - June 2023